A few notes...

- **HW7 (NASTRAN/FEMAP assignment) is due 11:59pm TONIGHT**

- **HW8 (textbook Problems 5.1-5.4) is officially assigned today and is due 11:59pm Mon Apr 13**

- **We will one more exam before the final, aiming to have it on Fri Apr 17, will finalize this by Fri Apr 10 (our next class period)**

- **After the exam we will spend our remaining class periods on Chapter 8 material, which will be "fair game" for the final**

5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- **In applying the GSFF, for each stringer, $\bar{y}$ and $\bar{z}$ will be the coordinates of the point where the stringer lies and *A'* will be the effective area of the stringer**

- **So beginning at point A we have:** $\Rightarrow$ $$q_{AB} - 0 = \left[-(105.7)(2") - (18.12)(-1.667")\right]\left(0.5\ in^2\right)$$

  $\Rightarrow$ $$\boxed{q_{AB} = -90.60\ \frac{lb}{in}}$$

- **At point B we have:** $\Rightarrow$ $$q_{BC} - \overbrace{\left(-90.60\ \frac{lb}{in}\right)}^{q_{AB}} = \left[-(105.7)(2") - (18.12)(2.333")\right]\left(0.3\ in^2\right) = -76.10$$

  $\Rightarrow$ $$\boxed{q_{BC} = -166.7\ \frac{lb}{in}}$$

5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- **At point C we have:**

$$\Rightarrow \quad q_{CD} - \overbrace{\left(-166.7\ \frac{lb}{in}\right)}^{q_{BC}} = \left[-(105.7)(-4")-(18.12)(2.333")\right]\left(0.2\ in^2\right)$$

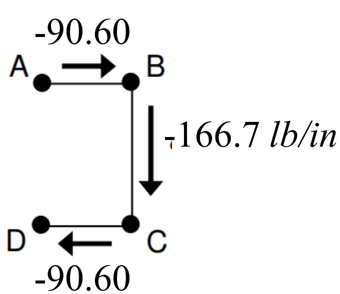
$$\Rightarrow \quad \boxed{q_{CD} = -90.59\ \frac{lb}{in}}$$

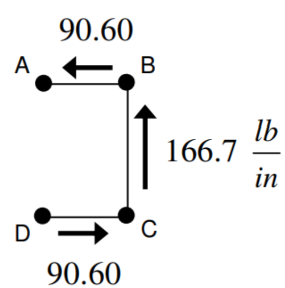
- **Note that we could solve this from the opposite end of the cross section as well:**

$$\Rightarrow \quad 0 - q_{CD} = \left[-(105.7)(-4")-(18.12)(-1{,}667")\right]\left(0.2\ in^2\right)$$

$$\Rightarrow \quad \boxed{q_{CD} = -90.60\ \frac{lb}{in}}$$

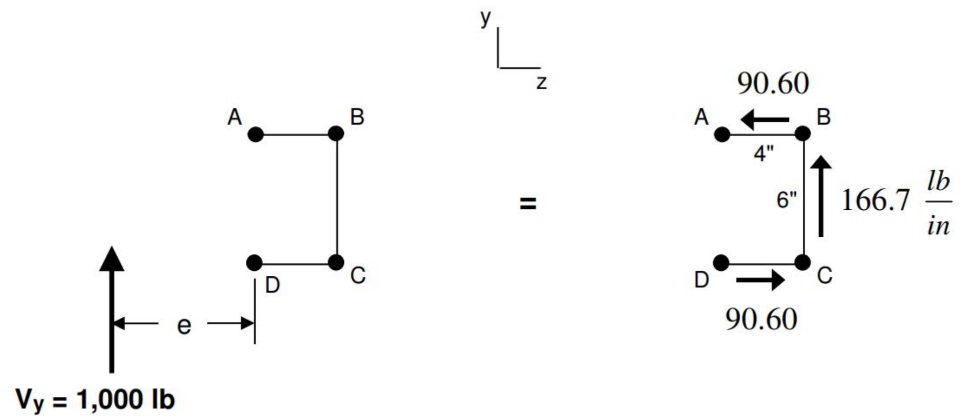
- **Thus we can draw our shear flow diagram in either of the following ways:**





5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- **We can do a check on our answers by performing force equivalence between the two diagrams below:**



<u>y-Direction</u>

$$\sum\left(F_y\right)_{\text{Left Figure}} = \sum\left(F_y\right)_{\text{Right Figure}}$$

$$1{,}000\ lb = \left(166.7\ \frac{lb}{in}\right)(6")$$

$$1{,}000 \approx 1{,}000.2 \quad \checkmark$$

DONE with Part (b)

<u>z-Direction</u>

$$\sum\left(F_z\right)_{\text{Left Figure}} = \sum\left(F_z\right)_{\text{Right Figure}}$$

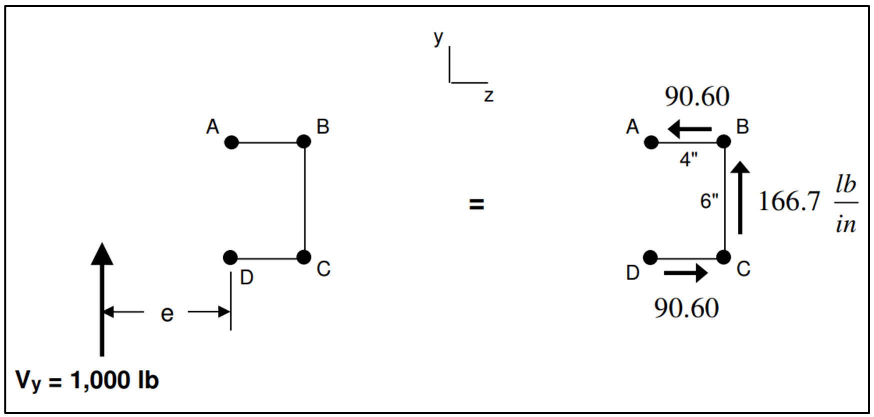
$$0 = -\left(90.60\ \frac{lb}{in}\right)(4") + \left(90.60\ \frac{lb}{in}\right)(4")$$

$$0 = 0 \quad \checkmark$$

5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- **Part (c) of this example says to find the location where the 1,000 lb load must act so that the beam will experience no torsion**

- **We can solve this by performing “moment equivalence” between the two diagrams below (calculating the moment about Point D):**

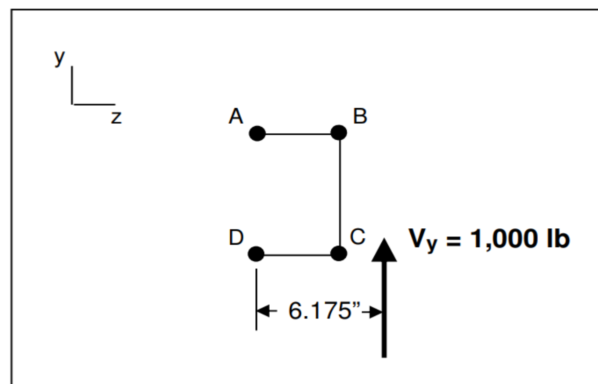


$$\circlearrowright + \quad \sum (M_D)_{\text{Left Figure}} = \sum (M_D)_{\text{Right Figure}}$$

$$-(1{,}000\ lb)\,e = \left[\left(90.60\ \frac{lb}{in}\right)(4")\right](6") + \left[\left(166.7\ \frac{lb}{in}\right)(6")\right](4") = 6175$$

$$\boxed{e = -6.175"}$$

- **The result is depicted in the following diagram**



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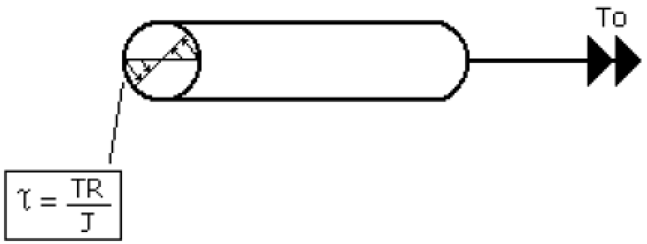
5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- **How do we know that this moment equivalence guarantees no torsion?**

- **Recall way back in Problem 1.1 (HW3), a beam was subjected to several loads, including a transverse force (V_y) and a torsional moment (T)**

- **As a result, the beam experienced internal shear stress due not only to V_y but also to T; these 2 components of shear stress were considered additive**

- **This torsional shear stress is depicted below:**



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- **The shear flow distribution we solved above assumed no torsion; therefore the total net moment about point D (or any point we choose) due to shear flow should equal the total net moment about point D (or any point we choose) due to V_y**

- **If V_y were acting anywhere other than 6.175" to the right of point D, the 2 moments would not be equal**

- **In order for the two moments to be equal in this case, there would have to be a component of torsional shear flow acting on the cross section in addition to the shear flow distribution we solved**

- **This would imply that V_y was causing torsion!**

DONE with Part (c)

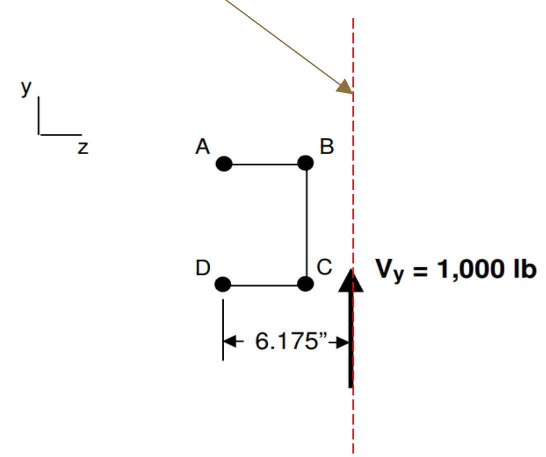
5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- **Finally, Part (d) of this example asks us to find the shear center**

- **This is the point on a beam cross-section at which the transverse force is applied such that the beam experiences bending but no twisting (i.e. torsion)**

- **It would seem that we already solved this above, however we have not yet considered a *z*-component of transverse force**

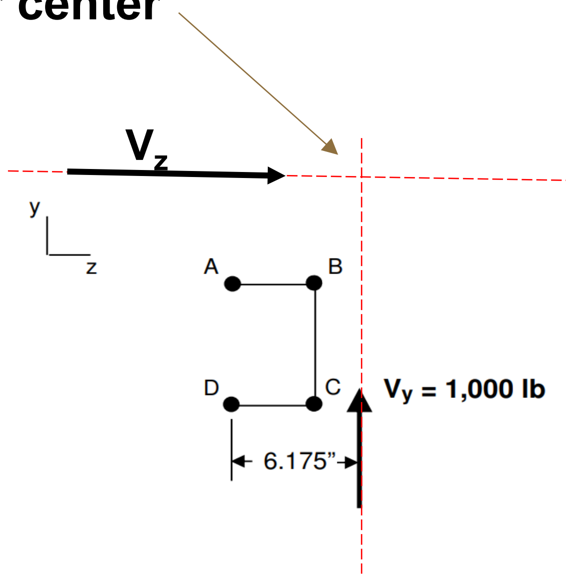
- **We know V_y could act anywhere along this vertical line and not cause torsion; but what if there were V_z involved as well?**



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- In that case we would be able to solve a location at which V_z would cause no torsion and draw a horizontal line through that location

- The intersection of these horizontal and vertical lines would then be the shear center

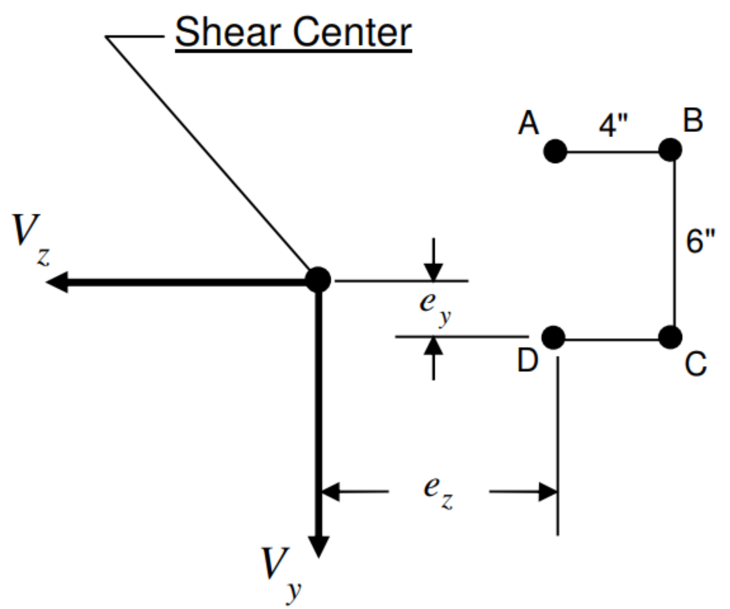


So how do we solve the location of the shear center?

5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- **First let's assume the shear center lies a distance e_y above point D and a distance e_z to the left of point D (with e_y and e_z yet to be determined)**

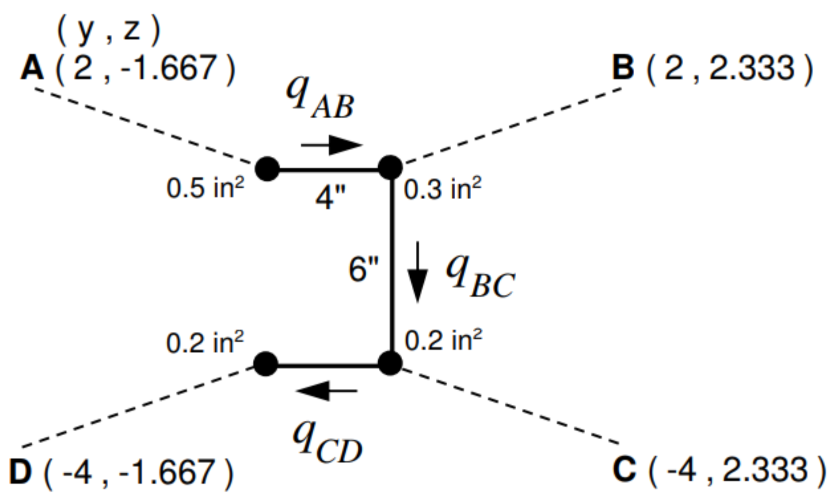
- **Assume there is a transverse force with both y and z components acting at the shear center:**



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- **We know for a given, V_y and V_z, we could use the GSFF to solve the shear flow distribution assuming no torsion**

- **This distribution would have the same general look as before (although for different values of q_{AB}, q_{BC}, and q_{CD}):**



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- **Let's use the GSFF to solve for q_{AB}, q_{BC}, and q_{CD} in terms of V_y and V_z:**

$$q_{out} - q_{in} = \frac{\left(I_y V_y - I_{yz} V_z\right)\overline{y}'A' + \left(-I_{yz} V_y + I_z V_z\right)\overline{z}'A'}{I_y I_z - I_{yz}^2}$$

- **Inserting the values for the inertias yields:**

$$q_{out} - q_{in} = \frac{\left\{\begin{aligned}&\left[\left(4.667\ in^4\right)V_y - \left(-0.8000\ in^4\right)V_z\right]\overline{y}'A' \\ &+\left[-\left(-0.8000\ in^4\right)V_y + \left(9.600\ in^4\right)V_z\right]\overline{z}'A'\end{aligned}\right\}}{\left(4.667\ in^4\right)\left(9.600\ in^4\right) - \left(-0.8000\ in^4\right)^2}$$

- **Simplifying yields:**

$$\boxed{q_{out} - q_{in} = \left\{\begin{aligned}&\left[\left(0.1057\right)\overline{y}' + \left(0.01811\right)\overline{z}'\right]V_y \\ &+\left[\left(0.01811\right)\overline{y}' + \left(0.2174\right)\overline{z}'\right]V_z\end{aligned}\right\}A'}$$

5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- **Following the same procedure as before to solve for q_{AB} yields:**

$$q_{out} = q_{AB} \qquad q_{in} = 0 \qquad A' = 0.5\ in^2 \qquad \bar{y}' = 2" \qquad \bar{z}' = -1.667"$$

$$\Rightarrow \qquad \boxed{q_{AB} = (0.09060)V_y - (0.1631)V_z}$$

- **Now solving for q_{BC} →**

$$q_{out} = q_{BC} \qquad q_{in} = q_{AB} \qquad A' = 0.3\ in^2 \qquad \bar{y}' = 2" \qquad \bar{z}' = 2.333"$$

$$\Rightarrow \qquad q_{BC} = q_{AB} + \left[(0.07610)V_y + (0.1630)V_z\right]$$

- **Inserting above expression for q_{AB} yields:** $\boxed{q_{BC} = (0.1667)V_y - (0.0001)V_z}$

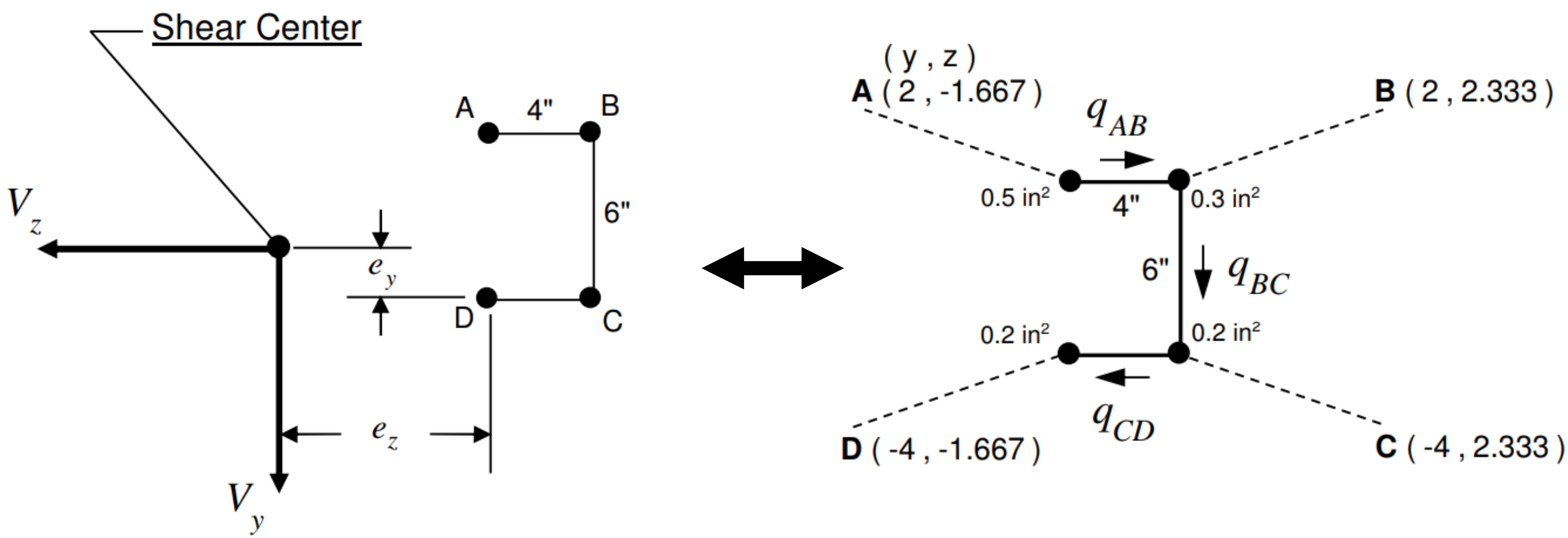
- **And solving for q_{CD} →**

$$q_{out} = 0 \qquad q_{in} = q_{CD} \qquad A' = 0.2\ in^2 \qquad \bar{y}' = -4" \qquad \bar{z}' = -1.667"$$

$$\Rightarrow \qquad \boxed{q_{CD} = (0.09060)V_y + (0.08700)V_z}$$

5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- **Now let's again perform "moment equivalence" between the two diagrams we've recently seen:**



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- **In particular, let's equate the moment in the *x* direction about point D:**

$$\circlearrowleft + \quad \sum \left(M_D\right)_{\text{Left Figure}} = \sum \left(M_D\right)_{\text{Right Figure}}$$

$$e_z V_y + e_y V_z = -\left[q_{AB}(4")\right](6") - \left[q_{BC}(6")\right](4")$$

- **Inserting the above expressions for q_{AB} and q_{BC} yields:**

$$e_z V_y + e_y V_z = \left\{ \begin{array}{l} -\overbrace{\left[(0.09060)V_y - (0.1631)V_z\right]}^{q_{AB}}(4")(6") \\ -\overbrace{\left[(0.1667)V_y - (0.0001)V_z\right]}^{q_{BC}}(6")(4") \end{array} \right\}$$

- **This simplifies to** $\boxed{\left(e_z + 6.175\right)V_y + \left(e_y - 3.916\right)V_z = 0}$

5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

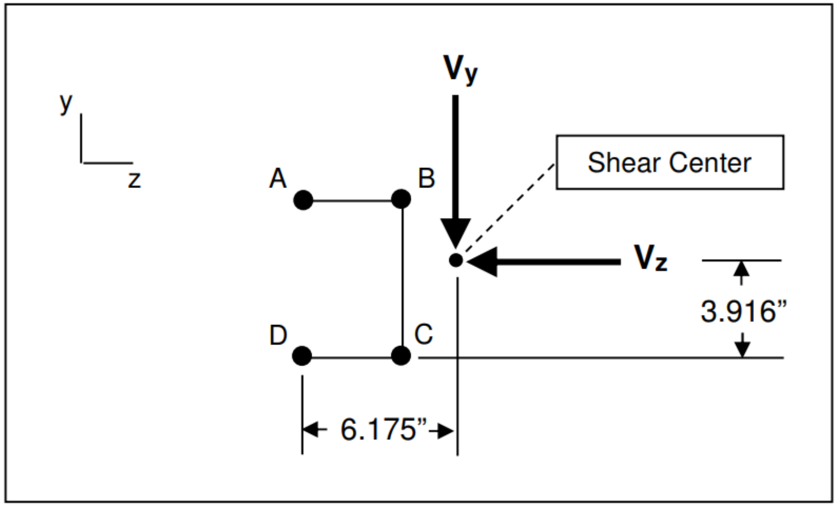
- **This equation must hold for any values of V_y and V_z, so each of the coefficients of V_y and V_z must = 0:**

$$\left(e_z + 6.175\right) = 0 \qquad \text{and} \qquad \left(e_y - 3.916\right) = 0$$

- **Thus the shear center is located at →**

$$\boxed{\begin{aligned} e_y &= 3.916" \\ e_z &= -6.175" \end{aligned}}$$

- **The diagram depicting the proper location of the shear center is as follows:**



DONE with Part (d)

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